RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Applied Sciences
Third Year - Semester II Examination - March-May 2026

STA 3402 - INTRODUCTION TO NEURAL NETWORKS AND MACHINE LEARNING

Time: Three (03) hours

Answer ALL (04) questions.
Write the answer to Question 1 on the table provided.
Provide R codes, outputs and interpretations for Q2-Q4.
Attach the question paper to the answer scripts.

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1. This question consists of Fifteen (15) Multiple Choice Questions. Write the correct answers in the appropriate places in the given table.

a)		f)		k)	
b)		g)		l)	
c)		h)		m)	
d)		i)		n)	
e)		j)		o)	

a) A Recurrent Neural Network (RNN) is particularly useful for which of the following tasks?

i. Classifying independent (non-sequential) data without temporal dependencies
ii. Predicting a sequence where past information influences future outputs
iii. Reducing the dimensionality of large datasets
iv. Clustering unlabeled data into groups
v. Performing linear regression on independent observations

b) A deep neural network trained for image classification shows very high training accuracy but poor test accuracy. Which of the following is the most appropriate explanation?

i. The model is underfitting due to insufficient complexity

ii. The model is overfitting due to excessive complexity

iii. The learning rate is exactly optimal

iv. The dataset has no noise

v. The activation functions are linear

c) During training of a deep neural network, the gradients become extremely small as they propagate backward through layers, slowing learning. Which technique is most suitable to address this issue?

i. Increasing the number of layers

ii. Using ReLU activation instead of sigmoid

iii. Reducing the dataset size

iv. Removing the loss function

v. Using only linear models

d) In a standard Feed-Forward Neural Network (FFNN) using backpropagation, which of the following statements is **true** regarding the "feed-forward" pass?

i. Weights are updated immediately after each layer processes the input data.

ii. Data flows from the input layer to the output layer without any cycles or loops.

iii. It uses the derivative of the loss function to adjust neurons in the hidden layers.

iv. Information travels backward from the output layer to the input layer to calculate the error.

v. Each layer's output is calculated by multiplying the current error gradient by the weight of the previous layer.

e) You are training a deep neural network and observe that the training loss oscillates heavily and does not converge. Which of the following is the most likely cause?

i. Learning rate is too high

ii. Model is too simple

iii. Dataset is perfectly balanced

iv. Activation functions are non-linear

v. Number of features is small

f) Consider a single neuron with input $x = 2$, weight $w = 0.5$, and bias $b = 1$. The activation function is *ReLU*. What is the output?

i. 0

ii. 1

iii. 2

iv. 3

v. 4

g) For a binary classification problem, the predicted probability is $\hat{y} = 0.8$ and the true label is $y = 1$. Using binary cross-entropy loss:

$$L = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

What is the approximate loss value?

i. 0.22

ii. 0.45

iii. 0.69

iv. 1.20

v. 2.30

h) Consider a neuron using the sigmoid activation function:

$$\sigma(z) = \frac{1}{1+\exp(-z)}$$

If $z = 0$, what is the output?

i. 0

ii. 0.25

iii. 0.5

iv. 0.75

v. 1

i) User A and User B have ratings on 3 items: $A : (5, 3, 0)$, $B : (4, 3, 1)$.

Using cosine similarity, what is the similarity between A and B?

i. 0.80

ii. 0.90

iii. 0.95

iv. 1.00

v. 0.60

j) Consider two sets: $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. What is the Jaccard distance between sets A and B?

i. 0.33

ii. 0.50

iii. 0.67

iv. 0.75

v. 1.00

k) Backpropagation is a learning technique that adjusts weights in the neural network by propagating weight changes,

i. Forward from inputs to outputs.

ii. Backward from outputs to inputs.

iii. Forward from inputs to hidden nodes.

iv. Backward from outputs to hidden nodes.

v. None of the above.

l) The number of nodes in the input layer is 10 and the hidden layer is 5. The maximum number of connections from the input layer to the hidden layer are,

i. 50

ii. 10

iii. More than 50

iv. In between 0 and 10

v. It is an arbitrary value

m) In a neural network, the Softmax function is most commonly used in which situation?

i. As an activation function for hidden layers

ii. In binary classification tasks

iii. To compute probabilities in multi-class classification tasks

iv. For reducing the dimensionality of data

v. In unsupervised learning tasks

For questions n) and o), use the following information.

A single neuron uses a sigmoid activation $\sigma(z) = \frac{1}{1+\exp(-z)}$. It has one input $x = 0.6$, weight $w = 0.5$, and bias $b = 0.1$. The learning rate is $\eta = 0.2$. The loss function, L is the mean squared error (MSE). For a single training example with target $t = 1$, after 1 **epoch** of training using gradient descent, calculate the following.

n) Value of the error derivative?

i. 1

ii. -0.5987

iii. -0.4013

iv. 0.4013

v. 0.5987

o) Value of the sigmoid derivative?

i. 1

ii. -0.5987

iii. 0.24

iv. 0.4013

v. 0.5987

2. a) Consider the given "consumer_data.txt" dataset. This represents a survey of 100 customers including their Age (numeric), Annual Spend (numeric), and Membership Type (categorical: Bronze, Silver, Gold).

   i. Extract the age and spend values for Observation 1 and Observation 2. Calculate the Euclidean Distance between these two points manually.

   (15 marks)

   ii. The Annual Spend variable has a much larger range than age. Explain how this difference in scale affects the Euclidean distance calculation and what step should be taken before clustering to ensure both variables contribute equally.

   (10 marks)

   b) A hospital data analyst has collected records on 300 patients from an oncology unit. The dataset is given in the "patient_data.txt" file. You will apply a technique to reduce dimensionality, then use the factor scores to perform a cluster analysis to identify patient subgroups.

   i. Explain, in no more than four sentences, why Principal Component Analysis (PCA) alone is not an appropriate dimension reduction technique for the patient dataset.

   (10 marks)

   ii. Fit a proper dimension reduction model retaining five dimensions. Display the scree plot and the percentage of variance explained by each dimension.

   (20 marks)

   iii. Consider the first two dimensions obtained from the dimension reduction result in part ii. Identify which variables drive each dimension and interpret what each dimension might represent clinically.

   (15 marks)

   iv. Extract a suitable number of individual factor scores from the dimension reduction results in part ii and perform agglomerative hierarchical clustering with Ward's linkage. Plot a dendrogram and use it to justify a choice of k clusters. Assign cluster labels to each patient.

   (30 marks)

3. Consider the following neural network set up. A fully-connected neural network with:

   - Two input neurons $x_1 = 1$, $x_2 = 0$
   - One hidden layer with two neurons h_1 and h_2, each using the **sigmoid** activation $\sigma(z) = \frac{1}{1+e^{-z}}$
   - One output neuron $\hat{y}$ with a **linear** (identity) activation
   - All biases fixed at $b = 1$
   - Weights as labelled:
     - Input to hidden: $w_1 = 0.5$, $w_2 = 1$, $w_3 = 0.8$, $w_4 = 0.3$

- Hidden to output: $v_1 = 1$, $v_2 = 0.6$

- Target output $y = 1$, loss function $L = \frac{1}{2}(y - \hat{y})^2$, learning rate $\eta = 0.5$

a) Draw the network architecture for the given set up. Label all nodes, weights, biases, activations, inputs, and the output.

(10 marks)

b) Perform the forward pass. Compute the pre-activation (net input) and post-activation output for each hidden neuron, then compute $\hat{y}$. Show all arithmetic.

(20 marks)

c) Compute the loss L. Then find $\frac{\partial L}{\partial \hat{y}}$.

(20 marks)

d) Using backpropagation, compute the gradients $\frac{\partial L}{\partial V_1}$ and $\frac{\partial L}{\partial V_2}$ (gradients of the loss with respect to the output-layer weights).

(20 marks)

e) Backpropagate through the hidden layer. Compute $\frac{\partial L}{\partial w_1}$ and $\frac{\partial L}{\partial w_2}$ (gradients with respect to the two weights connecting x_1 to the hidden layer). Show the sigmoid derivative step clearly.

(30 marks)

4. A Support Vector Machine (SVM) is a powerful supervised learning algorithm that is widely used in binary classification tasks to find the optimal boundary that best separates two classes.

a) Consider the following dataset.

X_1	X_2	Class
1	9	+
5	5	+
1	1	+
8	5	-
13	1	-
13	9	-

i. Plot the points in a plot. Clearly mark the two types of data points using + and -

ii. Sketch the support vectors (You may circle the corresponding data points)

iii. Sketch the decision boundary for a linear SVM classifier with maximum margin for this data set.

(15 marks)

b) Using R and setting the random seed to 123, split the dataset "banana_quality.csv" into 80% training and 20% testing sets. The dataset contains 8000 observations and 8 variables, where the final variable represents the quality of bananas (Good/Bad).

i. Perform appropriate data preprocessing and carry out exploratory data analysis (EDA) relevant to classifying the quality of bananas. Clearly describe the steps taken and interpret the key findings.

(20 marks)

ii. Develop Support Vector Machine (SVM) models to predict banana quality using the following kernels appropriately:

i. Linear kernel

ii. Radial Basis Function (RBF) kernel

Provide details of model fitting, including any tuning performed.

(40 marks)

iii. Evaluate the performance of both models using the test dataset:

i. Compute and compare test accuracy.

ii. Construct and interpret the confusion matrix for each model.

iii. Calculate precision, recall, and F1-score for both classes.

iv. Clearly identify the best-performing model, justifying your choice based on these metrics.

(25 marks)

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